

Economics-Aware Machine Learning for Option-Implied Risk Metrics

Yi Cao¹ Zexun Chen² Yuhan Cheng³ **Heqing Shi²**

¹Xi'an Jiaotong-Liverpool University

²University of Edinburgh

³Shandong University

05 Dec, 2025

Motivation

- ▶ **Intuition:** option prices encode investor expectations, risk preferences for the underlying asset returns. **Forward-looking in nature.**
- ▶ Decoding this information provides economic implications:
 - Tail risk management.
 - Moment-based factor construction.
- ▶ Benefits:
 - Free of historical data.
 - Free of statistical assumptions.
 - Incorporate market conditions rapidly.
- ▶ All the above require an estimation method to uncover the asset return density (or some distributional metrics) implied in option prices.

Literature Review

- ▶ Since the 1970s till the early 2000s, structural estimation methods were dominant, by using **parametric models**: Black and Scholes (1973); the CEV model by Cox and Ross (1976); the GARCH model by Duan (1995); the SV model by Heston (1993) and etc.
- ▶ Breeden and Litzenberger (1978): **model-free** risk-neutral density (RND) from option prices; Aït-Sahalia and Lo (1998): nonparametric kernel regression; Shimko (1993), Bliss and Panigirtzoglou (2004) propose to model IVs with constant extrapolation; Figlewski (2010) propose GEV extrapolation.

Literature Review

- ▶ Tail risk metrics. Aït-Sahalia and Lo (2000), Huggenberger et al. (2018), Barone-Adesi (2016): option-implied VaR, ES; Vilkov and Xiao (2013), Gao et al. (2019), Gao and Pan (2024): option-implied tail loss index.
- ▶ Moment factors. Bakshi et al. (2003), Dennis et al. (2006), Fu et al. (2016), Bali et al. (2019): option-implied volatility, skewness, kurtosis can explain stock returns.

The Gap

- ▶ However, the classical estimation methods of RND from option prices suffer from the dilemma—go parametric, or go model-free?
- ▶ Parametric:
 - Well-conditioned, smooth, analytical RNDs.
 - At the cost of rigid assumptions.
- ▶ Model-free:
 - Free of statistical assumptions; data-driven RNDs.
 - At the cost of no economic awareness; sensitive to noisy data.
- ▶ The problem for **stocks** is even severe, neither approaches work well so far.
- ▶ Can we model the noisy option prices nonparametrically, but with some basic economic principles?

Our Paper

- ▶ We propose a novel Constrained Gaussian Process (ConsGP), which
 - Can learn option prices nonparametrically.
 - And make predictions that align with the no-arbitrage principles.
- ▶ The ConsGP enables us to uncover the RND and the distributional metrics in a completely model-free way:
 - more suitable for stocks than the existing methods.
 - Both call and put options are used to uncover the RND.
 - A generic approach to derive option-implied distributional metrics, e.g., VaR, ES, partial moments, central moments, ...

Our Paper

- ▶ Empirical analysis with the options of a large cross-section of stocks.
- ▶ Stock options are noisier and more illiquid than index options, a challenge for both parametric and nonparametric methods.
- ▶ S&P 500 constituent stocks; options traded from 1996 to 2022.
- ▶ The stock-level RND recovered with ConsGP provides significantly enhanced economic information:
 - The implied left-tail risk metrics are **3.92x** better in terms of the out-of-sample R^1 ;
 - The implied moments are **4.30x** better in terms of the Sharpe ratio, compared with the classical model-free RND recovery methods.

Model

Model-Free RND Recovery

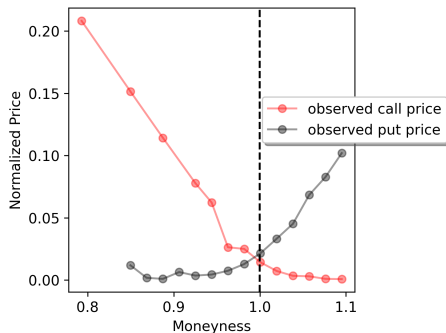
Theorem (Breedon and Litzenberger (1978))

For a specified expiry time T , at the current time t , if the prices of a cross-section of European options with different strikes satisfy the no-arbitrage principles, then there exists a risk-neutral distribution whose cdf is given by:

$$F^{\mathbb{Q}}(K) = e^{r(T-t)} \partial_K C_t(K) + 1, \text{ for call options,} \quad (1)$$

$$F^{\mathbb{Q}}(K) = e^{r(T-t)} \partial_K P_t(K), \text{ for put options.} \quad (2)$$

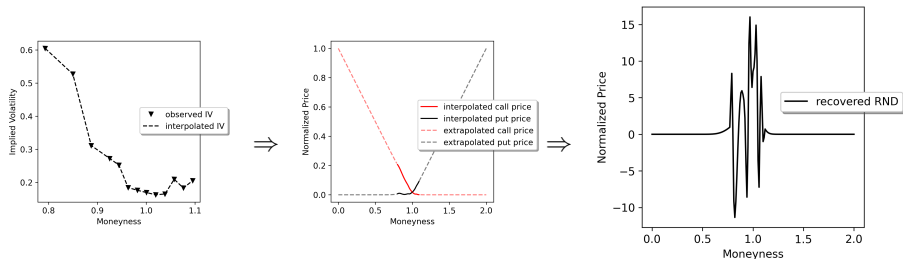
Model-Free RND Recovery



Traded option prices for CSCO on 28 Jun 2021 with 25 days to the expiration.

- ▶ Option prices in reality are not arbitrage-free, and are not traded on a continuum of strikes.
- ▶ The Breeden-Litzenberger theorem can not work!

Model-Free RND Recovery



IV interpolation with lognormal tails extrapolation.

- ▶ Classical recovery methods, e.g., Shimko (1993), Bliss and Panigirtzoglou (2004), Figlewski (2010), work with IVs.
- ▶ Spline interpolation on IVs, and parametric tail extrapolation.
- ▶ The model-free RND can still be ill-conditioned!

Economics-Aware ML: The ConsGP

- ▶ **Our approach:** we employ ML to model option prices directly, but with the constraints of no-arbitrage conditions. → An **economics-aware** machine.
- ▶ We use a Gaussian Process, a nonparametric probabilistic ML model.
- ▶ We perturb its predictive distribution by our prior knowledge.

Economics-Aware ML: The ConsGP

We consider a GP subject to the no-arbitrage prior knowledge:

$$y_m(x) = \underbrace{\sum_{j=1}^m \xi(t_j) \phi_j(x) + \epsilon}_{\text{finite-dimensional GP with noise}}, \text{ s.t. } \underbrace{\begin{cases} y_m(x_i) = y_i & \text{(interpolation conditions),} \\ y_m \in \mathcal{E} & \text{(inequality conditions),} \end{cases}}_{\text{fit to the observed prices + no-arbitrage constraints}}$$

where $\mathcal{E}$ is a convex set of $\mathbb{R}^m$ represents the no-arbitrage constraints:

$$\mathcal{E}_\kappa := \begin{cases} \forall j = 1, \dots, m : \overbrace{\mathcal{E}_j \geq 0}^{\text{positivity}} & \text{if } \kappa = 0, \\ \forall j = 2, \dots, m : \overbrace{\mathcal{E}_j - \mathcal{E}_{j-1} \leq 0}^{\text{non-increasing}} \text{ (call)}, \overbrace{\mathcal{E}_j - \mathcal{E}_{j-1} \geq 0}^{\text{non-decreasing}} \text{ (put)} & \text{if } \kappa = 1, \\ \forall j = 3, \dots, m : \overbrace{\mathcal{E}_j - \mathcal{E}_{j-1} \geq \mathcal{E}_{j-1} - \mathcal{E}_{j-2}}^{\text{convexity}} & \text{if } \kappa = 2, \end{cases}$$

- ▶ $t_1, \dots, t_m \in \mathbb{R}^+$ is a set of equally spaced knot points.
- ▶ $\phi_j(x)$ is a hat basis function.

Economics-Aware ML: The ConsGP

- ▶ The standard GP fits $y_m(x) = \Phi\xi + \epsilon$ to the data, where $\xi \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta)$, as there is no better knowledge about ξ . $\rightarrow$ Data-driven learning.
- ▶ From economics (the no-arbitrage conditions), we actually know more:

$$\xi \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta), \text{ s.t. } \begin{cases} \Phi\xi = y & \leftarrow \text{what we can learn from } \mathbf{data}, \\ \xi \in \mathcal{E} & \leftarrow \text{what we know from } \mathbf{economics}. \end{cases}$$

- ▶ How to solve the vector ξ then?

Economics-Aware ML: The ConsGP

- ▶ Step one: estimate a standard GP (interpolation conditions):

$$\mathcal{L}_m(\theta; \mathbf{X}, y) = -\frac{1}{2} \log(\det(\Sigma_\theta)) - \frac{1}{2} y_m^{obs\top} \Sigma_\theta^{-1} y_m^{obs} - \frac{n}{2} \log 2\pi,$$
$$\theta_{MLE} = \arg \min_{\theta \in \Theta} -\mathcal{L}_m(\theta; \mathbf{X}, y).$$

- ▶ Step two: evaluate the optimal posterior covariance $\Sigma_{\theta_{MLE}}$.

Economics-Aware ML: The ConsGP

- Step three: solve the ξ with no-arbitrage constraints:

$$\Lambda^{(1)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \Lambda^{(2)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \quad \Lambda^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \end{bmatrix}$$

- The no-arbitrage constraints are in $\Lambda = [\Lambda^{(1)}, \Lambda^{(2)}, \Lambda^{(3)}]^\top$
- ξ is no more a Gaussian, but a Gaussian with constraints:

$$\xi | \{ \Phi \xi = y, l \leq \Lambda \xi \leq u \} \leftarrow \text{the predictive distribution}$$

- Consider the mode of ξ , which can be solved by a QP:

$$\xi^* = \arg \min_{\xi \in \mathbb{R}^m} \underbrace{\xi^\top \Sigma_{\theta_{MLE}}^{-1} \xi}_{\propto \log \text{ Gaussian pdf}}$$

s.t. $\Phi \xi = y^{\text{obs}}, \leftarrow$ interpolation conditions, **data**

$l \leq \Lambda \xi \leq u, \leftarrow$ inequality conditions, **economics**

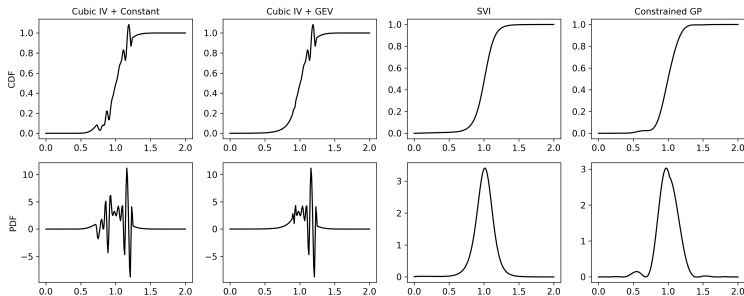
Economics-Aware ML: The ConsGP

- ▶ Step four: economics-aware predictions: $\hat{y}_m(\mathbf{X}) = \Phi_{\mathbf{X}}\xi^*$.
- ▶ We use ConsGP to predict the no-arbitrage call and put option prices on **any** strikes $\rightarrow$ Breeden-Litzenberger theorem applicable!
- ▶ Construct an RND concerning both call and put (the second derivative):

$$f^{\mathbb{Q}}(K) := \begin{cases} e^{r(T-t)} \frac{\partial^2 \hat{y}_m^{(\text{put})}(K; S_t, T)}{\partial K^2} & \text{if } K \leq S_t, \\ e^{r(T-t)} \frac{\partial^2 \hat{y}_m^{(\text{call})}(K; S_t, T)}{\partial K^2} & \text{if } K > S_t. \end{cases}$$

The ConsGP RND-Recovery

- ▶ Cubic IV + Constant: cubic-splined IV with constant extrapolation, i.e., log-normal tails, Shimko (1993), Bliss and Panigirtzoglou (2004).
- ▶ Cubic IV + GEV: cubic-splined IV with GEV tails, Figlewski (2010).
- ▶ SVI: parametrized IV, Gatheral (2004).
- ▶ ConstrainedGP: our method.



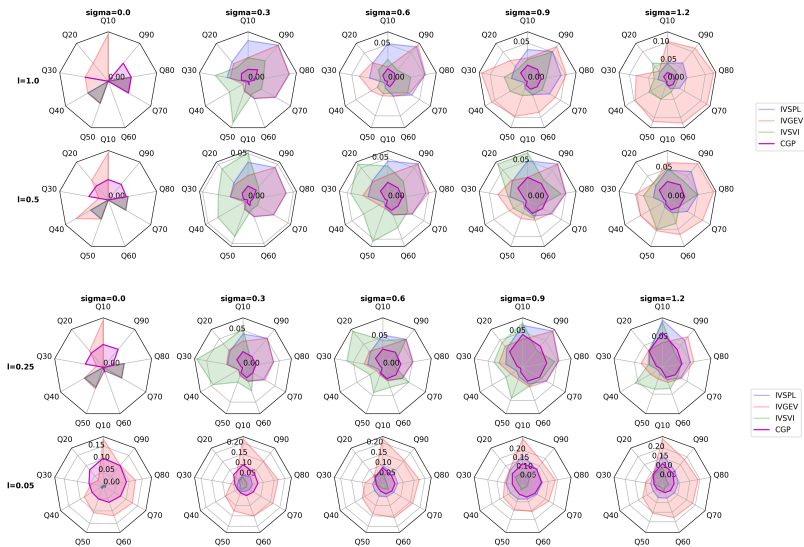
The RNDs recovered by different methods. COF, 05 Aug 2021, 71 days to the expiration.

Simulation

Simulations

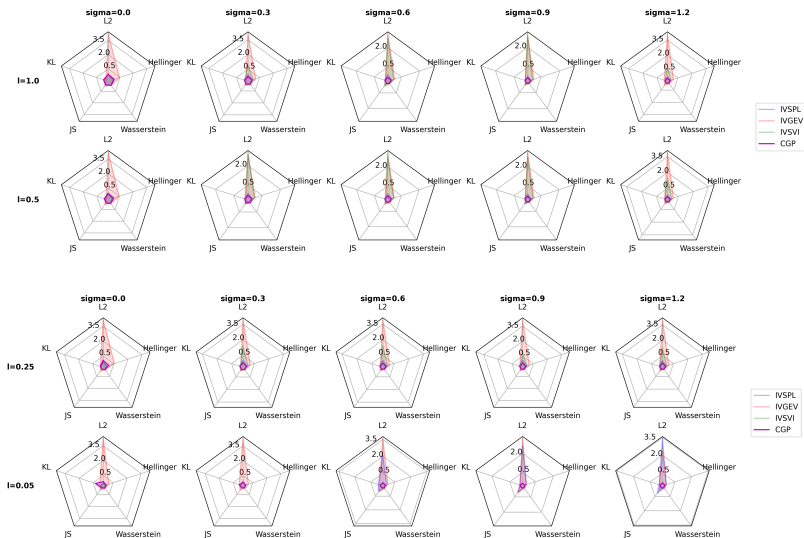
- ▶ Heston's model with fixed parameters. The “true” RND is known.
- ▶ Generate 100 call and 100 put prices on the equidistant strikes $K \in [20, 200]$.
- ▶ Sample 100%, 50%, 25% and 5% of the total 100 calls and puts.
- ▶ Add 0ϵ , 0.3ϵ , 0.6ϵ , 0.9ϵ and 1.2ϵ random noise, $\epsilon \sim \mathcal{N}(0, 1)$.

Simulation



The quantile losses of the 10%, 20%, ..., 80%, 90% percentiles.

Simulation



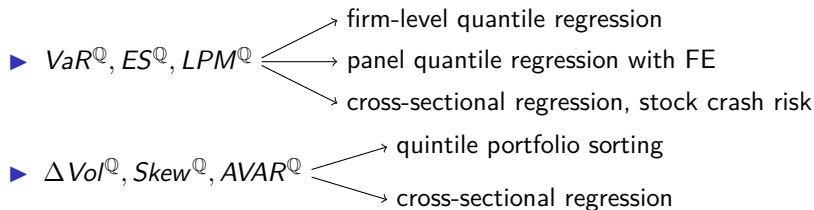
The divergence losses (KL, JS, Wasserstein, and Hellinger).

Empirical Analysis

Data

- ▶ Constituent stocks in the S&P 500 index from Jan 1996 to Dec 2022.
- ▶ Stock options, returns, and the risk-free rates are from OptionMetrics.
- ▶ We apply minimal stock option data filtering to retain the most information.

Our Empirical Analysis Road-map



Left-Tail Risk

	Not winsorized		Winsorized	
	Option	Stock	Option	Stock
Mean	2.59	-0.24	2.53	0.43
Std	7.33	14.02	5.12	4.56
Min	-58.86	-262.62	-11.52	-28.02
25%	-0.36	-1.21	-0.29	-1.18
50%	1.79	0.20	1.79	0.20
75%	4.77	2.32	4.65	2.23
Max	54.36	35.43	26.66	19.32
No. of firms	456	456	446	446

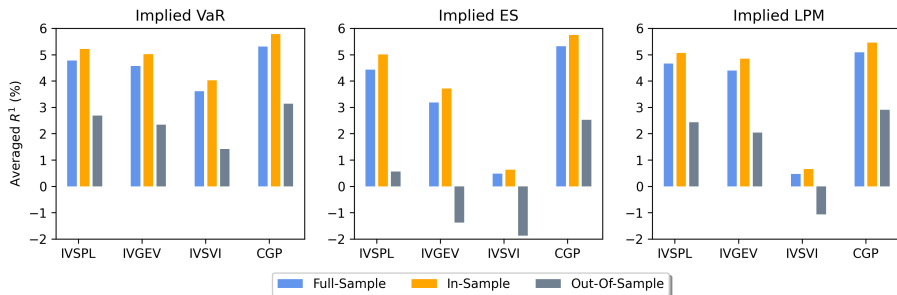
Out-of-sample quantile regression R^1 , averaged across all stocks.

Left-Tail Risk

	Option	Stock	Option+Stock	Firm-chara	BKM*	Idiosyn*	Raw
VaR^Q	-2.854*** (-18.79)		-2.826*** (-29.33)	-3.143*** (-36.01)	-3.092*** (-36.59)	-3.049*** (-4.84)	-2.254*** (-14.63)
VaR^P		-0.915*** (-18.26)	-0.023 (0.51)				
$Beta_{mkt}$				0.003*** (2.61)	0.004*** (3.00)	0.002 (0.22)	0.003*** (2.87)
Coskew				0.012*** (9.79)	0.011*** (8.87)	0.010 (1.00)	0.010*** (8.46)
Size				-0.006*** (-9.08)	-0.004*** (-6.04)	-0.003 (-0.69)	-0.003*** (-5.60)
Ivol							-0.065*** (-8.54)
$Ivol^\perp$					-0.059*** (-7.54)	-0.062 (-1.01)	
Iskew					0.000 (-1.07)	0.000 (-0.13)	0.000 (-1.09)
Ikurt					0.000 (0.94)	0.000 (0.12)	0.000 (1.00)
Idiovol							0.022*** (3.74)
$Idiovol^\perp$						0.022 (0.75)	
Idioskew						0.000 (0.30)	0.000*** (2.84)
Firm-Week Obs.	301,040	301,040	301,040	300,727	300,72	300,727	300,727

The panel quantile regression (with fixed effect).

Left-Tail Risk

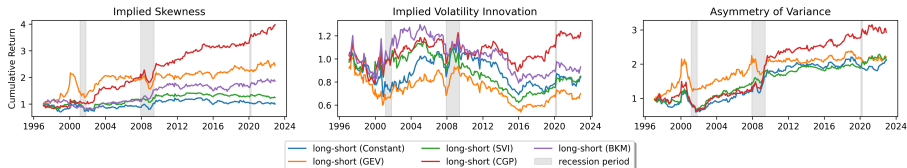


The averaged full-/ in-/ out-of-sample R^1 by different RND-recovery methods.

Moment-Based Factors

- ▶ Option-implied moment-based return factors from the literature:
 - Chordia et al. (2021), implied skewness
 - Dennis et al. (2006), $\Delta Vol \rightarrow Vol_t - Vol_{t-1}$
 - Bollerslev et al. (2020), Asymmetry of variance $\rightarrow UPM - LPM$
- ▶ We construct through ConsGP: $Skew^Q, \Delta Vol^Q, AVAR^Q$.

Moment-Based Factors



Long-short portfolios (quintile sorting).

Moment-Based Factors

	Constant		GEV		SVI		CGP	
	SR	<i>t</i> -Statistic	SR	<i>t</i> -Statistic	SR	<i>t</i> -Statistic	SR	<i>t</i> -Statistic
$Skew^Q$	0.02	0.29	0.12	1.76	0.05	0.71	0.20	2.9
ΔVol^Q	0.00	0.02	-0.023	-0.34	-0.00	-0.07	0.04	0.65
$AVAR^Q$	0.10	1.43	0.10	1.52	0.11	1.61	0.12	1.80

The Sharpe ratio and *t*-statistic of the long-short portfolios.

Moment-Based Factors

	$Skew^Q$				ΔVol^Q				$AVAR^Q$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
$\mathcal{M}_t$	0.001* (1.97)	0.001** (2.12)	0.001** (2.34)	0.001 (1.25)	0.021** (2.01)	0.019** (2.18)	0.019* (1.84)	0.020* (1.75)	5.328** (2.50)	5.566*** (3.29)	5.936*** (3.51)	5.459*** (2.91)
α_0	0.01*** (2.72)	0.003 (0.79)	0.005 (1.15)	0.001 (0.23)	0.009*** (3.03)	0.002 (0.46)	0.004 (0.73)	0.001 (0.09)	0.009*** (2.81)	0.004 (1.29)	0.006 (1.61)	0.004 (0.64)
Firm-chara	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
BKM	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes
Idiosyn	No	No	No	Yes	No	No	No	Yes	No	No	No	Yes
Firm-Month Obs.	60,597	60,442	60,401	50,207	48,639	48,609	48,609	48,609	60,282	60,128	60,090	50,076
adj. R^2	0.3%	8.1%	10.4%	12.0%	0.8%	8.6%	11.0%	12.2%	3.0%	9.8%	11.5%	13.1%

FM cross-sectional regressions. Monthly stock returns against lagged moment-based factors.

Moment-Based Factors

	$Skew^Q$				ΔVol^Q				$AVAR^Q$			
	Constant	GEV	SVI	CGP	Constant	GEV	SVI	CGP	Constant	GEV	SVI	CGP
$\mathcal{M}_t$	0.000 (1.02)	0.000 (1.44)	0.000 (0.04)	0.001* (1.97)	0.014 (0.81)	0.000 (-0.01)	-0.005 (-0.39)	0.021** (2.01)	3.364* (1.72)	1.103 (1.23)	0.124 (0.95)	5.328** (2.50)
α_0	0.01*** (2.94)	0.01*** (3.00)	0.01*** (3.10)	0.01*** (2.72)	0.009*** (3.08)	0.010*** (3.23)	0.009*** (2.78)	0.009*** (3.03)	0.009*** (3.00)	0.009*** (2.79)	0.009*** (2.82)	0.009*** (2.81)
Firm-Month Obs.	71,631	68,548	74,307	60,597	48,639	48,639	48,639	48,639	59,626	58,821	60,584	60,282
adj. R^2	0.7%	0.4%	0.0%	0.3%	1.2%	1.1%	0.7%	0.8%	1.94%	1.31%	0.71%	3.0%

FM cross-sectional regression (univariate). Factors are from different RND-recovery methods.

Conclusion

Conclusion

- ▶ We propose the ConsGP to nonparametrically model option prices:
 - No IVs, directly learn from option prices.
 - **Data-driven** with **economics-awareness** (the no-arbitrage conditions).
- ▶ The ConsGP RND-recovery produces well-conditioned RND.
- ▶ We examine the recovered RND information in a **stock-level** perspective.
- ▶ Our method enhances the economic information interpretable from options.

References I

- Aït-Sahalia, Y., Lo, A.W., 1998. Nonparametric estimation of state-price densities implicit in financial asset prices. *The Journal of Finance* 53, 499–547.
- Aït-Sahalia, Y., Lo, A.W., 2000. Nonparametric risk management and implied risk aversion. *Journal of Econometrics* 94, 9–51.
- Bakshi, G., Kapadia, N., Madan, D., 2003. Stock return characteristics, skew laws, and the differential pricing of individual equity options. *Review of Financial Studies* 16, 101–143.
- Bali, T.G., Hu, J., Murray, S., 2019. Option implied volatility, skewness, and kurtosis and the cross-section of expected stock returns. *Georgetown McDonough School of Business Research Paper* .
- Barone-Adesi, G., 2016. Var and cvar implied in option prices. *Journal of Risk and Financial Management* 9, 2.
- Black, F., Scholes, M., 1973. The pricing of options and corporate liabilities. *Journal of Political Economy* 81, 637–654.
- Bliss, R.R., Panigirtzoglou, N., 2004. Option-implied risk aversion estimates. *The Journal of Finance* 59, 407–446.
- Bollerslev, T., Li, S.Z., Zhao, B., 2020. Good volatility, bad volatility, and the cross section of stock returns. *Journal of Financial and Quantitative Analysis* 55, 751–781.
- Breeden, D.T., Litzenberger, R.H., 1978. Prices of state-contingent claims implicit in option prices. *Journal of Business* , 621–651.
- Chordia, T., Lin, T.C., Xiang, V., 2021. Risk-neutral skewness, informed trading, and the cross section of stock returns. *Journal of Financial and Quantitative Analysis* 56, 1713–1737.
- Cox, J.C., Ross, S.A., 1976. The valuation of options for alternative stochastic processes. *Journal of Financial Economics* 3, 145–166.

References II

- Dennis, P., Mayhew, S., Stivers, C., 2006. Stock returns, implied volatility innovations, and the asymmetric volatility phenomenon. *Journal of Financial and Quantitative Analysis* 41, 381–406.
- Duan, J.C., 1995. The garch option pricing model. *Mathematical finance* 5, 13–32.
- Figlewski, S., 2010. Estimating the Implied RiskNeutral Density for the US Market Portfolio. Oxford University Press. pp. 323–353.
- Fu, X., Arisoy, Y.E., Shackleton, M.B., Umutlu, M., 2016. Option-implied volatility measures and stock return predictability. *Journal of Derivatives* 24, 58–78.
- Gao, G.P., Lu, X., Song, Z., 2019. Tail risk concerns everywhere. *Management Science* 65, 3111–3130.
- Gao, J., Pan, J., 2024. Option-implied crash index. Available at SSRN 5035431 .
- Gatheral, J., 2004. A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives. Presentation at Global Derivatives & Risk Management, Madrid .
- Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies* 6, 327–343.
- Huggenberger, M., Zhang, C., Zhou, T., 2018. Forward-looking tail risk measures. *SSRN Electronic Journal* .
- Shimko, D., 1993. Bounds of probability. *Risk* 6, 33–37.
- Vilkov, G., Xiao, Y., 2013. Option-implied information and predictability of extreme returns. *SSRN Electronic Journal* .